

EXP-8 ASD Translation Scheme

Lex Code

Yacc Code

```

1 //{
2 #include<stdio.h>
3 #include<stdlib.h>
4 #include<string.h>
5
6 int yylex();
7 int yyerror(char*);
8 }
9
10 union{ int num; char *str; }
11 %token IF ELSE RE
12 %token <str> ID
13 %token <num> NUM
14
15 %%
16
17 S      :   IF '(' ID RE NUM ')'
18           { printf("\nEquivalent Switch Case: \n");
19             printf("switch(%s)\n{\n\n,$$);
20             printf("\tcase %d: ",$$);
21             }
22           STMT EL
23         ;
24
25 STMT    :   ID '=' NUM ';'
26           { printf("\n\t\t%s = %d;\n\t\t\tbreak;\n",$1,$3); };
27
28 EL      :   ELSE IF '(' ID RE NUM ')'
29             STMT EL
30           |
31             ELSE
32             STMT
33           ;
34 %%
35
36 int main()
37 {
38     printf("Enter the Statement in if-else: ");
39     yyparse();
40     return 0;
41 }
42 int yywrap()
43 {
44     return 1;
45 }
46 int yyerror(char *s)
47 {
48     printf("Error : %s\n",s);
49 }

```

Output:

```
mariy@Venkata-HP-Touch: /n x + v
mariy@Venkata-HP-Touch:/mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-8$
mariy@Venkata-HP-Touch:/mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-8$ lex 8.l
mariy@Venkata-HP-Touch:/mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-8$ yacc -d 8.y
mariy@Venkata-HP-Touch:/mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-8$ gcc lex.yy.c y.tab.c
mariy@Venkata-HP-Touch:/mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-8$ ./a.out
Enter the Statement in if-else: if(x==1) a = 5; else if (x==2) b = 10; else c = 20;

Equivalent Switch Case:
switch(x)
{
    case 1:
        a = 5;
        break;
    case 2:
        b = 10;
        break;
    default:
        c = 20;
        break;
}
mariy@Venkata-HP-Touch:/mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-8$ |
```